

Calculus Cheat Sheet

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Revision 4

Limits

Existence

$\lim_{x \rightarrow a} f(x) = L$ exists if $\forall \epsilon > 0 \exists \delta > 0$ s.t. $|x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$.

Properties

$$\begin{aligned}\lim_{x \rightarrow a} [cf(x)] &= c \lim_{x \rightarrow a} [f(x)] \\ \lim_{x \rightarrow a} [f(x) \pm g(x)] &= \lim_{x \rightarrow a} [f(x)] \pm \lim_{x \rightarrow a} [g(x)] \\ \lim_{x \rightarrow a} [f(x)g(x)] &= \lim_{x \rightarrow a} [f(x)] \lim_{x \rightarrow a} [g(x)] \\ \lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] &= \frac{\lim_{x \rightarrow a} [f(x)]}{\lim_{x \rightarrow a} [g(x)]}, \lim_{x \rightarrow a} [g(x)] \neq 0\end{aligned}$$

Evaluation Techniques

Basics

$\lim_{x \rightarrow a} [f(x)] = f(a)$ if f exists at a

L'Hôpital's Rule

If $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ then

$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \lim_{x \rightarrow a} \left[\frac{f'(x)}{g'(x)} \right]$$

Derivatives

Definition

$$\frac{d}{dx} f(x) = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Techniques

Sum Rule

$$\frac{d}{dx} (f(x) \pm g(x)) = f'(x) \pm g'(x)$$

Product Rule

$$\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

Quotient Rule

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Power Rule

$$\frac{d}{dx} (x^n) = nx^{n-1}$$

Chain Rule

$$\frac{d}{dx} (f(g(x))) = f'(g(x))g'(x)$$

Parametric

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ recurse y for n th derivative
 $(x, y) + (x', y')t$ for parametric tangent

Common Derivatives

$$\begin{aligned}\frac{d}{dx} (x) &= 1 \\ \frac{d}{dx} (\sin x) &= \cos x \\ \frac{d}{dx} (\cos x) &= -\sin x \\ \frac{d}{dx} (\tan x) &= \sec^2 x \\ \frac{d}{dx} (\sec x) &= \sec x \tan x \\ \frac{d}{dx} (\csc x) &= -\csc x \cot x \\ \frac{d}{dx} (\cot x) &= -\csc^2 x \\ \frac{d}{dx} (\sin^{-1} x) &= \frac{1}{\sqrt{1-x^2}} \sin^{-1} x \neq \frac{1}{\sin x} \\ \frac{d}{dx} (\cos^{-1} x) &= -\frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx} (\tan^{-1} x) &= \frac{1}{1+x^2} \\ \frac{d}{dx} (a^x) &= a^x \ln a \\ \frac{d}{dx} (e^x) &= e^x \\ \frac{d}{dx} (\ln x) &= \frac{1}{x}, x > 0 \\ \frac{d}{dx} (\ln |x|) &= \frac{1}{x}, x \neq 0 \\ \frac{d}{dx} (\log_a x) &= \frac{1}{x \ln a}\end{aligned}$$

Common Chain Rule Derivatives

$$\begin{aligned}\frac{d}{dx} ([f(x)]^n) &= n[f(x)]^{n-1} f'(x) \\ \frac{d}{dx} (e^{f(x)}) &= e^{f(x)} f'(x) \\ \frac{d}{dx} (\ln[f(x)]) &= \frac{f'(x)}{f(x)} \\ \frac{d}{dx} (\sin[f(x)]) &= f'(x) \cos[f(x)] \\ \frac{d}{dx} (\cos[f(x)]) &= -f'(x) \sin[f(x)] \\ \frac{d}{dx} (\tan[f(x)]) &= f'(x) \sec^2[f(x)] \\ \text{Trig derivatives, same old same old} \\ \frac{d}{dx} (f(x)^{g(x)}) &= \frac{g(x)f'(x)}{f(x)} + \ln[f(x)]g'(x)\end{aligned}$$

Integrals

Definition

The integral of some function $f(x)$ is a function $f^*(x)$ s.t. $f^{*'}(x) = f(x)$. **Don't forget your constant!** The integral can be done using the Right Riemann Sum:

$$\begin{aligned}\int_a^b f(x) dx &= \\ \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{b-a}{n} \cdot f\left(a + i \cdot \frac{b-a}{n}\right) \right]\end{aligned}$$

U-Substitution

$\int_a^b f(g(x)) \cdot g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$. Using U-Substitution, $u = g(x)$ and $du = g'(x)dx$ ($dx = \frac{du}{g'}$).

Integration By Parts

$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx$ and $\int_a^b u(x)v'(x) dx = uv|_a^b - \int_a^b u'(x)v(x) dx$.

Common Integrals

$$\begin{aligned}\int x^n dx &= \frac{1}{n+1} x^{n+1} + C \\ \int \frac{1}{x} dx &= \ln|x| + C \\ \int \frac{1}{ax+b} dx &= \frac{1}{a} \ln|ax+b| + C \\ \int \ln x dx &= x \ln x - x + C \\ \int e^x dx &= e^x + C \\ \int a^x dx &= \frac{a^x}{\ln a} + C \\ \int \sin x dx &= -\cos x + C \\ \int \cos x dx &= \sin x + C \\ \int \tan x dx &= \ln|\sec x| + C \\ \int \sec^2 x dx &= \tan x + C \\ \int \sec x \tan x dx &= \sec x + C \\ \int \csc x \cot x dx &= -\csc x + C \\ \int \csc^2 x dx &= -\cot x + C \\ \int \sec x dx &= \ln|\sec x + \tan x| + C \\ \int \frac{1}{a^2+x^2} dx &= \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C \\ \int \frac{1}{\sqrt{a^2+x^2}} dx &= \sin^{-1} \left(\frac{x}{a} \right) + C\end{aligned}$$

Trig Reduction Formulae

(For $n \geq 2$)

$$\begin{aligned}\int \sin^n(x) dx &= \\ -\frac{1}{n} \sin^{n-1}(x) \cos(x) + \frac{n-1}{n} \int \sin^{n-2}(x) dx \\ \int \cos^n(x) dx &= \\ \frac{1}{n} \cos^{n-1}(x) \sin(x) + \frac{n-1}{n} \int \cos^{n-2}(x) dx \\ \int \tan^n(x) dx &= \\ \frac{1}{n-1} \tan^{n-1}(x) + \frac{n-1}{n} \int \tan^{n-2}(x) dx \\ \int \sec^n(x) dx &= \frac{1}{n-1} \sec^{n-2}(x) \tan(x) + \\ \frac{n-2}{n-1} \int \sec^{n-2}(x) dx\end{aligned}$$

Trig Identities

$$\begin{aligned}\sin^2(x) + \cos^2(x) &= 1 \\ 1 + \tan^2(x) &= \sec^2(x) \\ \sin(2x) &= 2 \sin(x) \cos(x) \\ \cos(2x) &= \cos^2(x) - \sin^2(x) \\ &= 2 \cos^2(x) - 1\end{aligned}$$

$$= 1 - 2 \sin^2(x)$$

Trig Substitutions

$$\begin{aligned}\sqrt{a^2 - x^2} &\rightsquigarrow x = a \sin(\theta) \\ \sqrt{a^2 + x^2} &\rightsquigarrow x = a \tan(\theta) \\ \sqrt{x^2 - a^2} &\rightsquigarrow x = a \sec(\theta)\end{aligned}$$

Applications

Centroids

$$C = (\hat{x}, \hat{y}), \hat{x} = \frac{\int x f(x) dx}{\int f(x) dx} \ \& \ \hat{y} = \frac{\int y f(y) dy}{\int f(y) dy}$$

Arc Length

$$\begin{aligned}L &= \int_a^b \sqrt{1 + [f'(x)]^2} dx \\ L &= \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt\end{aligned}$$

for parametric. Area is just this times $2\pi x$ or y for opposite of axis.

Vector Calculus

Cartesian

$$d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}, d\tau = dxdydz$$

Gradient

$$\nabla t = \partial_x t \hat{\mathbf{x}} + \partial_y t \hat{\mathbf{y}} + \partial_z t \hat{\mathbf{z}}$$

Divergence

$$\nabla \cdot \mathbf{v} = \partial_x v_x \hat{\mathbf{x}} + \partial_y v_y \hat{\mathbf{y}} + \partial_z v_z \hat{\mathbf{z}}$$

Curl

$$\begin{aligned}\nabla \times \mathbf{v} &= (\partial_y v_z - \partial_z v_y) \hat{\mathbf{x}} \\ &+ (\partial_z v_x - \partial_x v_z) \hat{\mathbf{y}} \\ &+ (\partial_x v_y - \partial_y v_x) \hat{\mathbf{z}}\end{aligned}$$

Laplacian

$$\nabla^2 t = \partial_{xx} t + \partial_{yy} t + \partial_{zz} t$$

Spherical

$$d\mathbf{l} = dr\hat{\mathbf{r}} + r d\theta\hat{\boldsymbol{\theta}} + r \sin\theta d\phi\hat{\boldsymbol{\phi}},$$

$$d\tau = r^2 \sin\theta dr d\theta d\phi$$

Gradient

$$\nabla t = \partial_r t \hat{\mathbf{r}} + \frac{1}{r} \partial_\theta t \hat{\boldsymbol{\theta}} + \frac{1}{r \sin\theta} \partial_\phi t \hat{\boldsymbol{\phi}}$$

Divergence

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \frac{1}{r^2} \partial_r (r^2 v_r) \\ &+ \frac{1}{r \sin\theta} \partial_\theta (\sin\theta v_\theta) \\ &+ \frac{1}{r \sin\theta} \partial_\phi v_\phi \end{aligned}$$

Curl

$$\begin{aligned} \nabla \times \mathbf{v} &= \frac{1}{r \sin\theta} [\partial_\theta (\sin\theta v_\phi) - \partial_\phi v_\theta] \hat{\mathbf{r}} \\ &+ \frac{1}{r} \left[\frac{1}{\sin\theta} \partial_\phi v_r - \partial_r (r v_\phi) \right] \hat{\boldsymbol{\theta}} \\ &+ \frac{1}{r} [\partial_r (r v_\theta) - \partial_\theta v_r] \hat{\boldsymbol{\phi}} \end{aligned}$$

Laplacian

$$\begin{aligned} \nabla^2 t &= \frac{1}{r^2} \partial_r (r^2 \partial_r t) \\ &+ \frac{1}{r^2 \sin\theta} \partial_\theta (\sin\theta \partial_\theta t) \\ &+ \frac{1}{r^2 \sin^2\theta} \partial_\phi^2 t \end{aligned}$$

Cylindrical

$$d\mathbf{l} = ds\hat{\mathbf{s}} + s d\phi\hat{\boldsymbol{\phi}} + dz\hat{\mathbf{z}}, d\tau = s ds d\phi dz$$

Gradient

$$\nabla t = \partial_s t \hat{\mathbf{s}} + \frac{1}{s} \partial_\phi t \hat{\boldsymbol{\phi}} + \partial_z t \hat{\mathbf{z}}$$

Divergence

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \frac{1}{s} \partial_s (s v_s) \\ &+ \frac{1}{s} \partial_\phi (v_\phi) + \partial_z v_z \end{aligned}$$

Curl

$$\begin{aligned} \nabla \times \mathbf{v} &= \left[\frac{1}{s} \partial_\phi v_z - \partial_z v_\phi \right] \hat{\mathbf{s}} \\ &+ [\partial_z v_s - \partial_s v_z] \hat{\boldsymbol{\phi}} \\ &+ \frac{1}{s} [\partial_s (s v_\phi) - \partial_\phi v_s] \hat{\mathbf{z}} \end{aligned}$$

Laplacian

$$\nabla^2 t = \frac{1}{s} \partial_s (s \partial_s t) + \frac{1}{s^2} \partial_\phi^2 t + \partial_{zz} t$$

Vector Identities

$$\begin{aligned} \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) &= \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \\ \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \\ \text{etc..} &\text{ will fill in later} \end{aligned}$$

Theorems

Gradient Theorem

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla f) \cdot d\mathbf{l} = f(\mathbf{b}) - f(\mathbf{a})$$

Divergence Theorem

$$\int_V \nabla \cdot \mathbf{A} d\tau = \oint_S \mathbf{A} \cdot d\mathbf{a}$$

Curl (Stokes') Theorem

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint_C \mathbf{A} \cdot d\mathbf{l}$$

Coordinate Transforms

Spherical

$$\begin{cases} x = r \sin\theta \cos\phi \\ y = r \sin\theta \sin\phi \\ z = r \cos\theta \end{cases} \quad \begin{cases} \hat{\mathbf{x}} = \sin\theta \cos\phi \hat{\mathbf{r}} + \cos\theta \cos\phi \hat{\boldsymbol{\theta}} - \sin\phi \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{y}} = \sin\theta \sin\phi \hat{\mathbf{r}} + \cos\theta \sin\phi \hat{\boldsymbol{\theta}} + \cos\phi \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} = \cos\theta \hat{\mathbf{r}} - \sin\theta \hat{\boldsymbol{\theta}} \end{cases}$$

$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right) \\ \phi = \tan^{-1}\left(\frac{y}{x}\right) \end{cases} \quad \begin{cases} \hat{\mathbf{r}} = \sin\theta \cos\phi \hat{\mathbf{x}} + \sin\theta \sin\phi \hat{\mathbf{y}} + \cos\theta \hat{\mathbf{z}} \\ \hat{\boldsymbol{\theta}} = \cos\theta \cos\phi \hat{\mathbf{x}} + \cos\theta \sin\phi \hat{\mathbf{y}} - \sin\theta \hat{\mathbf{z}} \\ \hat{\boldsymbol{\phi}} = -\sin\phi \hat{\mathbf{x}} + \cos\phi \hat{\mathbf{y}} \end{cases}$$

Cylindrical

$$\begin{cases} x = s \cos\phi \\ y = s \sin\phi \\ z = z \end{cases} \quad \begin{cases} \hat{\mathbf{x}} = \cos\phi \hat{\mathbf{s}} - \sin\phi \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{y}} = \sin\phi \hat{\mathbf{s}} + \cos\phi \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} = \hat{\mathbf{z}} \end{cases}$$

$$\begin{cases} s = \sqrt{x^2 + y^2} \\ \phi = \tan^{-1}\left(\frac{y}{x}\right) \\ z = z \end{cases} \quad \begin{cases} \hat{\mathbf{s}} = \cos\phi \hat{\mathbf{x}} + \sin\phi \hat{\mathbf{y}} \\ \hat{\boldsymbol{\phi}} = -\sin\phi \hat{\mathbf{x}} + \cos\phi \hat{\mathbf{y}} \\ \hat{\mathbf{z}} = \hat{\mathbf{z}} \end{cases}$$